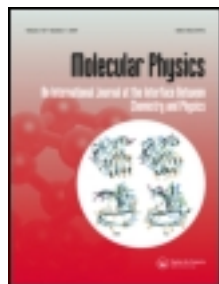


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Left–right correlation energy

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We first attempt to determine a local exchange functional $E_x[\rho]$ which accurately reproduces the Hartree–Fock (HF) energies of the 18 first and second row atoms. $E_x[\rho]$ is determined from ρ and $|\nabla\rho|$, and we find that we can improve significantly upon Becke's original generalized gradient approximation functional (commonly called B88X) by allowing the coefficient of the Dirac exchange term to be optimized (it is argued that molecules do not behave like the uniform electron gas). We call this new two parameter exchange functional OPTX. We find that neither $\nabla^2\rho$ or $\tau = \sum |\nabla\phi_i|^2$ improve the fit to these atomic energies. These exchange functionals include not only exchange, but also left–right correlation. It is therefore proposed that this functional provides a definition for exchange energy plus left–right correlation energy when used in Kohn–Sham (KS) calculations. We call this energy the Kohn–Sham exchange (or KSX) energy. It is shown that for nearly all molecules studied these KSX energies are lower than the corresponding HF energies, thus giving values for the non-dynamic correlation energy. At stretched geometries, the KSX energies are always lower than the HF energies, and often substantially so. Furthermore all bond lengths from the KSX calculations are longer than HF bond lengths and experimental bond lengths, which again demonstrates the inclusion of left–right correlation effects in the functional. For these reasons we prefer to split the correlation energy into two parts: left–right correlation energy and dynamic correlation energy, arguing that the usage of the words 'non-dynamic' or 'static' or 'near-degeneracy' is less meaningful. We recognize that this definition of KSX is not precise, because the definition of a local $E_x[\rho]$ can never be precise. We also recognize that these ideas are not new, but we think that their importance has been insufficiently recognized in functional determination. When we include third row atoms in our analysis, we are unable to find a local exchange functional which is a substantial improvement over B88X for the reproduction of HF energies. This must arise from the effects of the core orbitals, and therefore we do not consider that this detracts from the improved accuracy of OPTX. We report some MCSCF calculations constructed from bonding–antibonding configurations, from which we attempt to calculate *ab initio* left–right correlation. There is only moderate agreement between the two approaches. Finally we combine the OPTX functional with established correlation functionals (LYP, P86, P91) to form OLYP, OP86 and OP91; OLYP is a great improvement on BLYP for both energy and structure, and OP86, OP91 are an improvement over BP86, BP91 for structure. The importance of the exchange functional for molecular structure is therefore underlined.

1. Introduction

In quantum chemistry the determination of the correlation energy remains an extremely difficult and important problem. One generally accepted approach is that one works hard to obtain as near to the Hartree–Fock (HF) limit as possible. Then one performs some form of configuration interaction, either Møller–Plesset (MP) perturbation theory, or coupled-cluster (CC) calculations. Both of these approaches determine two types of electron correlation, without a separation into what are called the short-range part and the long-range part. The short-range part, commonly called dynamic correlation,

is concerned with the reduction in the value of the wavefunction as two electrons approach one another, arising from the r_{ij}^{-1} term in the hamiltonian. Many refer to this as arising from the effects of the electron–electron cusp. The shape of the cusp demands a linear behaviour of the wavefunction in r_{ij} (for unlike spins), and this means that all calculations are very slowly convergent with basis set. On the other hand we must observe that the shape of the cusp near $r_{ij} = 0$ is the same for every such electron pair, and therefore there is a huge amount of wasted effort in the construction of wavefunctions which separately introduce this special shape for every electron pair. The long-range part may be understood as that electron correlation effect which makes electrons go to separate atoms as the molecule dissociates. It is often

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called non-dynamic correlation or static correlation or near-degeneracy correlation.

In quantum chemistry the correlation energy is defined [1] as the energy difference $E(\text{HF}) - E(\text{Exact-non-relativistic})$. Near molecular equilibrium the correlation energy is nearly all dynamic (and may be estimated from $0.04 E_h$ per electron pair and $0.004 E_h$ for all unpaired electrons) and we shall see that the non-dynamic correlation energy is small. As the molecule dissociates, the HF approximation degrades, the correlation energy must therefore increase, the dynamic correlation must decrease because the number of electron pairs reduces, and therefore the non-dynamic correlation energy sharply increases. This discussion underlines the unsatisfactory importance of the HF method in quantum chemistry.

The alternative approach in quantum chemistry for the determination of the energy of a molecule is the multi-configurational self-consistent field (MCSCF) or its valence space variant, the complete active space (CASSCF) method. Both of these are designed to find the non-dynamic correlation energy through the introduction of valence space configurations into the configuration interaction (CI), although they necessarily introduce substantial dynamic correlation as well, because the effects cannot be separated. Then quantum chemists attempt to introduce the outstanding dynamic correlation, either through perturbation theory

(CASPT2) [2] or by CI (MCSCF-CI) [3]. The selection of the configurations to be included in the MCSCF is always a difficult decision.

As a summary there are three well-defined energy points in the quantum chemistry of a molecule: (i) the Hartree energy (H), which means all exchange effects turned off, (ii) the HF energy and (iii) the exact energy (Ex).

In density functional theory (DFT) we shall see that the situation is completely different.

2. Exchange functionals

For the present we shall discuss exchange-correlation functionals which depend upon the density $\rho(\mathbf{r})$, its gradient $\nabla\rho$, its laplacian $\nabla^2\rho$ and its so-called kinetic energy density $\tau = \sum_i (\nabla\phi_i)^2$. In other words we discuss local functionals $E_{xc}[\rho]$. It is usual to separate these into exchange $E_x[\rho]$ plus correlation $E_c[\rho]$. We shall in particular be examining $E_x[\rho]$.

The major part of $E_x[\rho]$ is the uniform electron gas expression due to Dirac (commonly called the LDAX functional)

$$E_x^{\text{LDA}} = - \sum_{\sigma} \left(\frac{3}{4} \right) \left(\frac{6}{\pi} \right)^{1/3} \int \rho_{\sigma}^{4/3}(\mathbf{r}) d\mathbf{r}. \quad (1)$$

In table 1 we show selected energies of the first and second row atoms. It is seen that LDAX finds 90% of the energy difference between the H and HF energies.

Table 1. Energies of first and second row atoms, calculated at the Hartree (H) level, HF level and KS(LDAX) level, in hartrees; the differences between HF and KS ($\Delta E = E(\text{KS}) - E(\text{HF})$) are given for B88X and OPTX.

Atom	HF	H	KS(LDAX)	$\Delta E(\text{B88X})$	$\Delta E(\text{OPTX})$
H	-0.5	-0.2419	-0.4571	+0.0021	+0.0013
He	-2.8617	-1.9510	-2.7236	-0.0017	-0.0022
Li	-7.4328	-5.7856	-7.1934	+0.0040	+0.0000
Be	-14.5730	-12.0622	-14.2233	+0.0066	+0.0000
B	-24.5293	-20.9866	-24.0663	+0.0034	+0.0011
C	-37.6900	-32.9053	-37.1121	+0.0014	+0.0019
N	-54.4045	-48.1569	-53.7093	+0.0036	+0.0005
O	-74.8140	-67.0178	-73.9975	-0.0165	-0.0066
F	-99.4114	-89.8509	-98.4754	-0.0329	-0.0148
Ne	-128.5471	-116.9881	-127.4907	-0.0430	-0.0265
Na	-161.8589	-148.3816	-160.6442	-0.0245	-0.0118
Mg	-199.6146	-184.1614	-198.2488	-0.0174	-0.0050
Al	-241.8768	-224.3680	-240.3573	-0.0103	-0.0007
Si	-288.8546	-269.1681	-287.1809	-0.0025	+0.0010
P	-340.7192	-318.7257	-338.8885	+0.0086	+0.0008
S	-397.5065	-373.1640	-395.5205	+0.0084	-0.0030
Cl	-459.4829	-432.6647	-457.3418	+0.0104	-0.0088
Ar	-526.8175	-497.3871	-524.5174	+0.0179	-0.0158
Mean Abs. $\Delta E/E_h$			1.075	0.0118	0.0057
Mean Abs. $\Delta E/\text{kcal mol}^{-1}$			674.46	7.40	3.56

The major improvement is the introduction of Becke's gradient correction [4]:

$$\Delta E_{\text{x}}^{\text{Becke88}} = - \sum_{\sigma} \int \frac{\beta \rho_{\sigma}^{4/3} x_{\sigma}^2}{1 + 6\beta x_{\sigma} \sinh^{-1} x_{\sigma}} \text{d}\mathbf{r}, \quad (2)$$

where $x_{\sigma} = |\nabla \rho_{\sigma}| \rho_{\sigma}^{-4/3}$. The introduction of this exchange functional followed earlier work by Becke [5] where he introduced his $X_{\alpha\beta\gamma}$ functional

$$E_{X_{\alpha\beta\gamma}} = E_{\text{x}}^{\text{LDA}} - \sum_{\sigma} \int \frac{\beta \rho_{\sigma}^{4/3} x_{\sigma}^2}{1 + \gamma x^2} \text{d}\mathbf{r}, \quad (3)$$

with $\beta = 0.0036$ and $\gamma = 0.004$. He commented at this early stage that 'the least squares error of this fit (to HF exchange energies) is surprisingly small'. He also observed that the optimum value of the parameter β was almost independent of Z . Becke had two important arguments to justify the form of his correction, equation (2): (i) in the asymptotic region the exchange energy density had the correct $-1/r$ form (although it later transpired that this form does not take over until one is very far from the molecule) and (ii) the parameter β was determined such that the computed energies of He, Ne, Ar, Kr, Xe and Rn agreed as closely as possible with the HF values. In fact Becke calculated the energies of these atoms perturbatively using LDA densities; his best value for β was 0.0042. In table 1 we give the corresponding energies calculated with the self-consistent KS procedure with this functional, commonly denoted B88X. It is seen that 99% of the energy difference between H and HF is now obtained.

Becke also showed that his functional gives excellent predictions (of a similar accuracy) for all atoms of the first three rows, which must be related to the fact that atoms only have dynamic correlation (i.e. they have zero non-dynamic correlation). We have looked for improved exchange functionals to reproduce the HF energies of these atoms. We have considered all 18 ground-state atoms of the first two rows (H–Ar). We have minimized

$$\Omega = \sum_{i=1}^{18} (T_s[\rho_i] + V_{\text{ne}}[\rho_i] + J[\rho_i] + \sum_j c_j \int F_j(\rho_i, \nabla \rho_i, \nabla^2 \rho_i, \tau_i) \text{d}\mathbf{r} - E_{\text{HF}}(i))^2. \quad (4)$$

For the expansion functionals F_j , we have used a number of trial forms.

- (A) The local density Dirac form, LDAX.
- (B) The Becke correction, equation (2), ΔB88X .
- (C) The form $\rho^{4/3} u^p$, where $u = \gamma x^2 / (1 + \gamma x^2)$, the term introduced by Becke for his 1986 functional [5], with various values of p . We used Becke's value for $\gamma = 0.004$.
- (D) The form $\rho^{4/3} \gamma y^2 / (1 + \gamma y^2)$, with $y = \nabla^2 \rho / \rho^{5/3}$.
- (E) The form $\rho^{4/3} \gamma \tilde{\tau} / (1 + \gamma \tilde{\tau})$, with $\tilde{\tau} = \sum_i (\nabla \phi_i)^2 / \rho^{5/3}$.

We performed unrestricted Kohn–Sham calculations using various combinations of the functional forms (A)–(E). We used the very large atomic gaussian basis sets of Partridge [6]. The corresponding values for Ω are given in table 2. The first calculation (i) using the LDAX functional is treated as a baseline. That it is so small demonstrates the excellence of LDAX as the starting approximation. The second calculation (ii) is B88X, and the great reduction of Ω demonstrates the great improvement achieved by this generalized gradient exchange (GGAX) functional. Calculation (iii) allows the coefficients of LDAX and ΔB88X to vary, the result being that Ω does not lower significantly. In calculation (iv) we use the alternative GGA form introduced by Becke in 1986 [5]. We see that this two-term form, involving $\rho^{4/3}$ and $\rho^{4/3} u^2$ gives a significant reduction in Ω . Calculations (v) and (vi) show that nothing significant is achieved through the introduction of u^1 or u^4 terms (although the u^4 does a little, we are not inclined to use it at the risk of numerical instability and overfitting of the data). Calculation (vii) is typical of our investigations to see if an inclusion of $\nabla^2 \rho$ lowers Ω , the result was negative. Likewise calculation (viii) is

Table 2. The determination of optimum local exchange functionals using various expansion forms (A)–(E); the table gives the coefficients c_j , equation (4), and the value for Ω , equation (4).

Functional	Expansion form	Coefficients	Ω
(i)	A	1.0	6.936×10^{-1}
(ii)	A, B	1.0, 1.0	4.888×10^{-3}
(iii)	A, B	1.0066, 0.9241	3.214×10^{-3}
(iv)	A, C($p = 2$)	1.0564, 2.3201	1.294×10^{-3}
(v)	A, C($p = 2$), C($p = 1$)	1.0468, 1.8442, 0.1617	1.140×10^{-3}
(vi)	A, C($p = 2$), C($p = 4$)	1.0519, 3.0524, -2.0213	8.664×10^{-4}
(vii)	A, C($p = 2$), D	1.0664, 2.4174, -0.1579	1.204×10^{-3}
(viii)	A, C($p = 2$), E	1.0527, 2.7589, -2.8582	7.089×10^{-4}

typical of our investigations to introduce τ , although the value of Ω is significantly reduced, discrepancies with the HF values for heavier atoms (see later) and the positive ions are very poor. We also computed functional (iv) re-optimizing γ . This gave the most significant reduction, with $\gamma = 0.006$ and $\Omega = 7.575 \times 10^{-4}$. This is therefore our new exchange functional.

In conclusion therefore, we have found a simple GGA functional which is a significant improvement over B88X, as judged by our criterion to give the HF energies of these atoms. It is

$$E_x[\rho] = \sum_{\sigma} \int - (1.05151 \times C_X + 1.43169 \times u_{\sigma}^2) \times \rho_{\sigma}^{4/3} d\mathbf{r}, \quad (5)$$

with $u = \gamma x^2 / (1 + \gamma x^2)$, with $\gamma = 0.006$, and C_X is the Dirac coefficient $(3/4)(6/\pi)^{1/3}$. We denote this new exchange functional as OPTX. As chemists it does not concern us unduly that LDAX does not have its theoretical value 1.0, the uniform electron gas is most unlike small atoms and molecules. For an atom there are no regions where the reduced density gradient x is near zero [7] (not even in the core, where the limiting value is $(4\pi)^{1/3}$ provided that a Slater basis is used). In the valence region $3.0 \leq x \leq 10.0$ and then x grows exponentially in the outer region. Molecules are made up of atoms, and therefore x will not be small, with one exception, and that is at the centre of a bond, where the density is a minimum. For H_2 , we find that $x < 0.25$ only for points within $0.07 a_0$ of the centre of the bond, but that y has the value -0.77 at the centre. The important point is that at the centre of the bond, where $x = 0.0$, $\nabla^2 \rho$ is not zero, and therefore the electron density is not constant locally. We conclude that a consideration of small x values is not necessary in order to obtain a good exchange functional. We are surprised that the inclusion of other powers of u , in particular the linear power, did not significantly improve the fit. We were not too surprised that inclusion of $\nabla^2 \rho$ did not improve matters, there have been a number of investigations which indicate that this is the case. We were more surprised that inclusion of τ terms did not significantly improve the fit (although it does a little), in the light of various claims in the literature. On the other hand we were pleased that this was the case, because we are looking for a functional form which gives a straightforward multiplicative exchange potential; we know from the work of those that generate optimized effective potentials (OEP) [8] that such local numerical multiplicative potentials give orbitals exceedingly close to Hartree-Fock orbitals. In summary we have not been able to find a significant improvement over our simple GGA form OPTX.

In table 1 we give the energies of the 18 atoms at the Hartree (H) level, the HF level (UHF, using our basis set), the LDAX level (i), the B88X level (ii), and the OPTX level (iv). We observe that (a) the results for OPTX are a significant improvement over B88X, and the greatest discrepancy for OPTX is for Ne ($-0.0265 E_h$). Furthermore the discrepancy from HF has reduced from B88X to OPTX for 16 of the 18 atoms (the very minor exceptions being He and C). The mean absolute discrepancy from the HF value has dropped from $7.40 \text{ kcal mol}^{-1}$ for B88X to $3.56 \text{ kcal mol}^{-1}$ for OPTX. We also found with OPTX, equally good agreement for the corresponding atomic positive ions. A similar table of values for B88X has been given by Adamo *et al.* [9], although the reason for the table was not explained.

We suggest that an important conclusion from this discussion is that an expansion of the 'exchange hole' (see below) to determine appropriate forms in an expansion for $E_x[\rho]$ does not seem beneficial; the above results suggest that the exchange energies of these atoms are very well represented by the above local forms for E_x . Indeed the introduction of some mix of 'exact' exchange into exchange-correlation functionals [10] does not seem justified on these arguments.

We have examined the predictions of OPTX for heavier atoms. We soon discovered that the OPTX energy for Kr is $0.81 E_h$ below the corresponding HF value! We have looked at the performance of OPTX for K, Ca, Mn, Cu, Zn, Ga, Ge, As, Se, Br and Kr and observe that the energy agreement steadily degrades from Ar and K to Kr (in particular the difficulty is not due to the presence of d electrons). We are unsure of the reason for this, but it has been known from the $X\alpha$ days that LDAX (for which $\alpha = 2/3$) is a much better approximation for heavier atoms than for lighter atoms (for which a larger value of α was preferred). Therefore inclusion of the heavier atoms makes the LDAX coefficient much closer to unity. This may explain why B88X performs much better than OPTX for these atoms (as reported by Becke [4] originally). Indeed Becke [5] stressed the importance of the universality of his first exchange functional. The largest difference for B88X is $0.31 E_h$ for Cu. We have carried out fits as above over 46 atoms (18 first and second row neutrals, 17 positive ions and 11 third row atoms), and conclude that we are unable to substantially improve over B88X. We have to conclude that the high Z values of the third row atoms require a different local exchange functional to the first two rows, and that it may not be possible to get a good local exchange functional for all three rows, if the total exchange energy is to be fit. However we consider that only the valence exchange energy is relevant to

chemistry, and we therefore expect OPTX to be equally good for molecules containing third row atoms.

3. Molecular calculations with exchange functionals

Whilst these exchange functionals produce KS energies which are very close to the HF energies of atoms, the same is not true for molecules. In table 3 we examine some energies of the N_2 molecule, obtained using a PVQZ basis. In particular HF and KS (using B88X and OPTX) energies are given at R_e , $1.5R_e$, $2.0R_e$ (here we arbitrarily define R_e as the HF value, $R_e = 1.0655 \text{ \AA}$). It is seen that the KS energies, using B88X, are lower than the HF energies by 0.0831, 0.1628 and 0.2482 E_h respectively. Thus without question the B88X functional cannot be regarded as an approximation to HF exchange, it must be introducing some correlation as well. Another way of seeing this is to optimize the geometry of N_2 using the HF and KS methodology. From table 3 we see that the HF bond length is 1.0655 $\text{\AA}$ (0.0322 $\text{\AA}$ too short) and the KS(B88X) bond length is 1.1121 $\text{\AA}$ (0.0144 $\text{\AA}$ too long). The KS bond lengths are 0.04 $\text{\AA}$ longer than the HF values, indicating again the presence of correlation effects in the functional.

These exchange functionals do not include dynamic correlation. Dynamic correlation is like that for correlation energy in a uniform electron gas, found numerically by Ceperley and Alder [11], and parametrized by Vosko *et al.* [12]. It is also apparently well represented by the Lee–Yang–Parr functional [13] (with parameters determined from the correlation energy of two electron atoms). Because of the construction of the Kohn–Sham procedure, exchange–correlation functionals must also include an aspect of the kinetic energy ($T[\rho] - T_s[\rho]$), we think that this is usually within dynamic correlation functionals. It is found that the addition of exchange functionals to correlation functionals give quality exchange–correlation functionals (e.g. B+LYP= BLYP).

Local exchange functionals must therefore introduce non-dynamic correlation. Now we must understand this.

- (i) The H_2 molecule at infinite separation has a HF energy which lies above $-1.0 E_h$. The two term CI calculation σ_g^2, σ_u^2 gives the correct dissociation energy, and therefore the HF energy is too high by an amount equal to the two electron exchange integral ($\sigma_g \sigma_u | \sigma_g \sigma_u$). At infinite separation, H_2 has zero dynamic correlation. Thus the non-dynamic correlation energy is an exchange integral. We deduce that we cannot separate exchange and non-dynamic correlation.
- (ii) There has been much discussion about the exchange hole, and this is relevant to our under-

standing. It is defined by writing the exchange energy as

$$E_x = -\frac{1}{2} \sum_{\sigma} \iint \frac{\rho_{\sigma}(\mathbf{r}_1) \rho_{x\sigma}(\mathbf{r}_1, \mathbf{r}_2)}{r_{12}} d\mathbf{r}_1 d\mathbf{r}_2, \quad (6)$$

where the exchange hole is defined in terms of the one density $\rho_{1\sigma}(\mathbf{r}_1, \mathbf{r}_2)$ and the density $\rho(\mathbf{r})$

$$\rho_{x\sigma}(\mathbf{r}_1, \mathbf{r}_2) = |\rho_{1\sigma}(\mathbf{r}_1, \mathbf{r}_2)|^2 / \rho_{\sigma}(\mathbf{r}_1). \quad (7)$$

In fact it is only the spherically averaged exchange hole which is needed, obtained by integrating over a sphere of radius s centred on $\mathbf{r}_1$, for $\mathbf{r}_2$:

$$\rho_{x\sigma}(\mathbf{r}_1, s) = \frac{1}{4\pi} \int \rho_{x\sigma}(\mathbf{r}_1, \mathbf{r}_1 + \mathbf{s}) d\Omega_s. \quad (8)$$

This spherically averaged exchange hole has been looked at by many workers, in particular in the HF approximation and in DFT. In particular if one studies H_2 at a large separation, with the reference electron $\mathbf{r}_1$ near proton A, in the HF approximation the exchange hole contains half an electron near A and half an electron near the other proton B. In this case $\rho_{x\sigma}(\mathbf{r}_1, \mathbf{r}_2) = \rho_{\sigma}(\mathbf{r}_1 + \mathbf{s})$ and

$$\int 4\pi s^2 \rho_{x\sigma}(\mathbf{r}_1, s) ds = 1, \quad (9)$$

that is, the reference electron is exchanging with half an electron near A and half an electron near B (with only the effects of half an electron contributing to the exchange integral). This is what is wrong, the reference electron should be exchanging with one electron near A. In the local density approximation (LDA) the exchange hole is localized near the reference electron $\rho_{x\sigma}(\mathbf{r}_1, s) = 9\rho_{\sigma}^2(k_F s)(k_F s)^{-2}$, and therefore the difficulty is not present. The local nature of this exchange functional means that there are no exchange effects arising from the electron density near B, which is correct. Therefore the KS energy using the LDAX functional is lower than the HF energy at large separations. The Becke–Roussel exchange functional [14] is based on a local representation of the exchange hole; it involves $\rho, \nabla\rho, \nabla^2\rho$ and the kinetic density. The corresponding exchange hole can be precisely calculated, and it can be numerically demonstrated [7] that it contains one electron near A, for the above example. These arguments must hold for all local exchange functionals.

In quantum chemistry, configuration interaction must remove the HF exchange error. For the H_2 example, it formally does through the introduction of the σ_u^2 configuration. It is well established that this configuration introduces left–right electron correlation. It is appropriate at this stage to comment on the importance of distinguishing between left–right correlation and correlation effects which arise from near degeneracy. In table

Table 3. Left-right correlation energies (in E_h) evaluated with the exchange functionals B88X and OPTX, calculated at the optimized HF geometries, and also at 1.25 and 1.5 magnification of HF geometries. Optimized geometries ($\text{\AA}$, $^\circ$), calculated with HF, LDAX, B88X and OPTX are also given. The HF and LDAX energies are reported.

Molecule	Parameter	HF	LDAX	B88X	OPTX	(Equil.)
H ₂	E	-1.1335	-1.0431	-0.0016	0.0031	
	HH	0.7338	0.7818	0.7522	0.7510	0.7414
H ₂ (1.25R _e (HF))	E	-1.1166	-1.0368	0.0033	0.0078	—
H ₂ (1.5R _e (HF))	E	-1.0831	1.0131	0.0095	0.0134	—
N ₂	E	-108.9945	-107.7488	0.0831	0.0711	—
	NN	1.0655	1.1056	1.1121	1.1115	1.0977
N ₂ (1.25R _e (HF))	E	-108.8499	-107.6446	0.1628	0.1482	—
N ₂ (1.5R _e (HF))	E	-108.6307	-107.5300	0.2482	0.2275	—
O ₂	E	-149.6920	-148.2240	0.1201	0.1020	—
	OO	1.1578	1.2198	1.2473	1.2313	1.2075
O ₂ (1.25R _e (HF))	E	-149.5942	-148.1822	0.2004	0.1720	—
O ₂ (1.5R _e (HF))	E	-149.4492	-148.1005	0.2723	0.2366	—
CO	E	-112.7905	-111.5251	0.0760	0.0605	—
	CO	1.1019	1.1381	1.1466	1.1464	1.1283
CO(1.25R _e (HF))	E	-112.6733	-111.4581	0.1356	0.1186	—
CO(1.5R _e (HF))	E	-112.5592	-111.3764	0.1742	0.1545	—
FH	E	-100.0683	-99.1443	0.0542	0.0396	—
	HF	0.8969	0.9432	0.9405	0.9310	0.9169
FH(1.25R _e (HF))	E	-100.0284	-99.1261	0.0755	0.0573	—
FH(1.5R _e (HF))	E	-99.9619	-99.0781	0.0955	0.0746	—
P ₂	E	-681.4960	-677.9119	0.0185	0.0307	—
	PP	1.8525	1.9176	1.9393	1.9276	1.8934
P ₂ (1.25R _e (HF))	E	-681.3946	-677.8604	0.0794	0.0846	—
P ₂ (1.5R _e (HF))	E	-681.2519	-677.7691	0.1405	0.1390	—
S ₂	E	-795.1057	-791.2139	0.0115	0.0373	—
	SS	1.8664	1.9281	1.9655	1.9386	1.8892
S ₂ (1.25R _e (HF))	E	-795.0234	-791.1767	0.0657	0.0809	—
S ₂ (1.5R _e (HF))	E	-794.9116	-791.0978	0.1163	0.1254	—
Si ₂	E	-577.7836	-574.4716	0.0133	0.0083	—
	SiSi	2.2185	2.3063	2.3370	2.3182	2.246
Si ₂ (1.25R _e (HF))	E	-577.6994	-574.4475	0.0798	0.0702	—
Si ₂ (1.5R _e (HF))	E	-577.6561	-574.4014	0.0829	0.0690	—
CH ₂ (³ B ₁)	E	-38.9404	-38.3022	-0.0071	-0.0007	—
	CH	1.0693	1.1062	1.0953	1.0945	1.08
	HCH	131.84	135.17	133.39	132.67	134.0
CH ₂ (³ B ₁)(1.25R _e (HF))	E	-38.8766	-38.2646	0.0142	0.0191	—
CH ₂ (³ B ₁)(1.5R _e (HF))	E	-38.7693	-38.1837	0.0412	0.0431	—
CH ₂ (¹ A ₁)	E	-38.8955	38.2603	0.0018	0.0023	—
	CH	1.0946	1.1453	1.1337	1.1358	1.11
	HCH	103.72	100.42	100.77	100.08	102.0
CH ₂ (¹ A ₁)(1.25R _e (HF))	E	-38.8376	-38.2314	0.0271	0.0274	—
CH ₂ (¹ A ₁)(1.5R _e (HF))	E	-38.7402	-38.1634	0.0571	0.0556	—
H ₂ O	E	-76.0658	-74.2454	0.0452	0.0382	—
	OH	0.9396	0.9839	0.9807	0.9733	0.958
	HOH	106.23	104.21	103.75	103.29	104.5
H ₂ O(1.25R _e (HF))	E	-75.9901	-75.2081	0.0828	0.0699	—
H ₂ O(1.5R _e (HF))	E	-75.8647	-75.1178	0.1216	0.1041	—
NH ₃	E	-56.2238	-55.4812	0.0221	0.0243	—
	NH	0.9980	1.0381	1.0338	1.0299	1.012
	HNH	107.97	106.22	105.61	104.90	106.7
NH ₃ (1.25R _e (HF))	E	-56.1188	-55.4222	0.0658	0.0624	—
NH ₃ (1.5R _e (HF))	E	-55.9452	-55.2956	0.1179	0.1094	—
CH ₄	E	-40.2164	-39.5290	-0.0067	0.0021	—

continued

Table 3. *Continued.*

Molecule	Parameter	HF	LDAX	B88X	OPTX	(Equil.)
	CH	1.0815	1.1144	1.1074	1.1059	1.0860
CH ₄ (1.25R _e (HF))	E	-40.0893	-39.4484	0.0346	0.0403	-
CH ₄ (1.5R _e (HF))	E	-39.8701	-39.2795	0.0872	0.0879	-
O ₃	E	-224.3752	-222.2814	0.2579	0.2263	-
	OO	1.1924	1.2708	1.3126	1.2892	1.272
	OOO	119.39	118.21	118.16	118.47	116.8
O ₃ (1.25R _e (HF))	E	-224.2257	-222.2367	0.4029	0.3504	-
O ₃ (1.5R _e (HF))	E	-223.9117	-222.0209	0.5355	0.4711	-
C ₂ H ₂	E	-76.8558	-75.4570	0.0355	0.0345	-
	CC	1.1791	1.2145	1.2168	1.2179	1.203
	CH	1.0540	1.0894	1.0772	1.0775	1.062
C ₂ H ₂ (1.25R _e (HF))	E	-76.6671	-75.6309	0.1067	0.1042	-
C ₂ H ₂ (1.5R _e (HF))	E	-76.3622	-75.4038	0.1904	0.1818	-
C ₂ H ₄	E	-78.0699	-76.8748	0.0125	0.0189	-
	CC	1.3136	1.3390	1.3500	1.3484	1.334
	CH	1.0738	1.1107	1.1009	1.1003	1.081
	HCC	121.65	121.67	121.86	121.84	121.3
C ₂ H ₄ (1.25R _e (HF))	E	-77.8541	-76.7379	0.0905	0.0925	-
C ₂ H ₄ (1.5R _e (HF))	E	-77.5031	-77.4733	0.1867	0.1807	-

1 it is seen that the exchange functional OPTX reproduces the HF energy of Be to within $0.0000 E_h$, whereas a two configuration ($1s^2 2s^2$, $1s^2 2p^2$) MCSCF calculation gives an energy $0.0179 E_h$ lower than the HF value. Thus the exchange functional does not include this near degeneracy effect. This can be understood, because it is easy to show that such a configuration mixing introduces r_{ij}^2 , which is a dynamic correlation effect.

All our arguments demonstrate that the use of a local exchange functional necessarily introduces left-right correlation into the energy expression. We argue again that the introduction of 'exact' exchange cannot be helpful in this context, because it reintroduces some of the above described HF error. To put the same argument another way, we do not believe that it can be correct to determine local exchange functions by adjusting their parameters to give good fits to molecular HF energies because local exchange functionals always introduce left-right correlation. Even though such fits may be numerically possible at equilibrium, at stretched geometries the fits must severely degrade. Voorhis and Scuseria [15] determine exchange functionals through a fitting to molecular HF energies.

We therefore prefer to separate the correlation energy into two parts called 'left-right' correlation and 'dynamic' correlation. We prefer to replace the usual terminologies 'non-dynamic' or 'static' or 'near-degeneracy', which are not precise, by the more precise 'left-right'. We therefore believe that in DFT, the key energies are the Hartree (H) energy, the local exchange limit

(which is HF exchange plus left-right correlation), which we call KSX, and the exact energy (Ex).

Gritsenko *et al.* [16] (GSB) have also recognized that GGA exchange does not approximate HF exchange for molecules. We quote '... GGA exchange energies are consistently too negative ... It is argued that GGA exchange functionals represent effectively not only exchange, but also molecular non-dynamical correlation'. They reach this conclusion after studying Li₂, N₂, F₂ and H₂. They have computed high accuracy *ab initio* densities, and from these computed KS orbitals, they then compute the exchange two-electron integral expression $-\sum_{ij}(ij|ij)$, which at equilibrium geometries is close to the HF value. For F₂, N₂ and stretched H₂, they show that this expression lies substantially above the B88X exchange value, by amounts similar to those presented in table 3. They recall that Slater [17] hoped that the exchange functional might describe both exchange and left-right correlation. They go on to estimate values for the non-dynamical correlation energy by computing CI energies for wavefunctions which dissociate correctly, and then show that these energy values are quite close to the B88X values. By similarly computing dynamic correlation energies ($E_{\text{corr}} - E_{\text{nd}}$), they show that these values are close to the values obtained from the LYP functional. In conclusion GSB were aware that GGA exchange introduced left-right correlation, and as they say, this concept has been around for many years. Thus the basic premise of our paper is certainly well established. However GSB appear uncertain of the consequences with statements such as

'GGA exchange energies are consistently too low'. It is our view that GSB are too closely wedded to the idea that exchange must be represented by $-\sum_{ij}(ij|ij)$. We think that the introduction of the two-electron exchange integrals into the theory or into the interpretation removes a distinct beauty of DFT. We stress again that 'exchange' and 'left-right correlation' cannot be separated.

4. Exchange calculations on molecules

In table 3 we report the results of simple studies on small molecules using a PVQZ basis set. Optimized energy calculations are reported using HF, LDAX, B88X and OPTX functionals at equilibrium HF geometries (R_e) and stretched HF geometries ($1.25R_e$, $1.5R_e$, angles unchanged). Optimized geometries are also reported, and compared with 'observed' values. It is readily verified that use of the LDAX values from tables 1 and 3 predict substantial molecular overbinding, a fact which is well known.

First we see that all calculated HF bond lengths are shorter than observed bond lengths, whereas all calculated B88X and OPTX bond lengths are longer than observed values. The difference between HF and OPTX bond lengths is substantial. These observations are explained by observing that HF is missing left-right correlation (introduction of dynamic correlation makes the HF bond lengths even shorter!), and OPTX is missing dynamic correlation effects. We also find that the OPTX bond lengths are shorter than the B88X bond lengths.

The energy differences $E(\text{HF})-E(\text{B88X})$ and $E(\text{HF})-E(\text{OPTX})$ are also given. These correspond to our definition of left-right correlation energy. We first observe that the B88X energies are lower than the OPTX energies, e.g. for N_2 by $0.0120 E_h$, and for O_2 by $0.0181 E_h$, although it must be noted that for the N and O atoms, $E(\text{B88X})-E(\text{OPTX})$ is 0.0031 and $-0.0099 E_h$ respectively.

We now look at $E(\text{HF})-E(\text{OPTX})$. At all stretched geometries these differences are positive, and often substantially so. For $\text{N}_2(1.5R_e)$, $\text{O}_2(1.5R_e)$ and $\text{O}_3(1.5R_e)$, the $E(\text{HF})-E(\text{OPTX})$ values are 0.2275 , 0.2366 and $0.4711 E_h$. The correlation energy introduced by this functional for these molecules is substantial! It is greatest for multiple bonded systems. At the HF equilibrium structures, the values are much smaller and can be slightly negative (CH_2). We expect indeed that left-right correlation should be smaller at the more compact structures.

The overall conclusion from the results in table 3 is that the energies from GGA KS calculations using local exchange-only functionals, do not approximate to HF calculations on these molecules. The values for the

OPTX left-right correlation energies differ from those obtained with B88X.

5. *Ab initio* calculations

We have performed some *ab initio* calculations with which to compare our DFT calculations for values of left-right correlation. Essentially all that we can do is to perform MCSCF [3] calculations which include all double excitations generated by replacing bonding orbitals by their corresponding antibonding orbitals. We also include quadruple excitations etc which comprise simultaneous bonding-antibonding excitations. We have to recognize that such calculations will introduce more than left-right correlation. For example, for H_2 , with only the two configurations σ_g^2 and σ_u^2 , at short internuclear distances these become s^2 and p^2 , in which case angular (i.e. dynamic) correlation is introduced. We must therefore expect that the MCSCF energies can be lower than the OPTX energies. Results of these calculations are presented in table 4, where the third column gives the *ab initio* correlation energies, computed as $E(\text{HF})-E(\text{MCSCF})$, and the fourth column gives the DFT left-right correlation energies, computed with OPTX as $E(\text{HF})-E(\text{OPTX})$ (corrected by the amount by which OPTX does not exactly reproduce the HF atomic energies, as given in table 1). The active bonding orbitals are given in the second column, the corresponding antibonding orbitals are included in the MCSCF calculations.

The results in table 4 show interesting observations. It is best to compare the *ab initio* correlation energies with the 'corrected' values in the final column. For H_2 the *ab initio* values are greater than the GGAX values, as expected from the above arguments. For FH (with only the $3\sigma^2 \rightarrow 4\sigma^2$ configuration included), H_2O and CH_4 , the *ab initio* values are smaller than the GGAX values, for reasons which are not clear. For N_2 , the agreement between the two approaches is good. For ozone the results are interesting: both approaches show substantial left-right correlation at the stretched geometries (although different), but at equilibrium the *ab initio* value is much smaller; perhaps some configurations in the CI have been omitted. Overall it appears difficult to get agreement between the two ways of obtaining left-right correlation energy.

6. A new exchange-correlation functional

We have remarked earlier on the usual separation of the exchange-correlation functional into a sum of separate exchange and correlation parts $E_{xc}[\rho] = E_x[\rho] + E_c[\rho]$. The main finding of this paper, to this stage, is that we may have found an improved exchange functional OPTX. It is therefore appropriate to examine combinations of this functional with existing correlation

Table 4. MCSCF energy calculations, using a PVQZ basis, at HF(R_e), and stretched HF (1.25 R_e , 1.5 R_e) geometries, giving a comparison of *ab initio* non-dynamic correlation energies with local exchange KS left-right correlation energies.

Molecule	Active space	E(HF)–E(MCSCF)	E(HF)–E(OPTX)
N ₂ (R_e)	3 σ_g 1 π_u	0.0700	0.0721
N ₂ (1.25 R_e)	3 σ_g	0.1272	0.1492
N ₂ (1.5 R_e)	3 σ_g	0.2155	0.2285
FH(R_e)	3 σ	0.0227	0.0261
FH(1.25 R_e)	3 σ	0.0340	0.0438
FH(1.5 R_e)	3 σ	0.0500	0.0611
O ₃ (R_e)	6a ₁ 4b ₂ 1b ₁	0.0452	0.2063
O ₃ (1.25 R_e)	6a ₁ 4b ₂ 1b ₁	0.1837	0.3304
O ₃ (1.5 R_e)	6a ₁ 4b ₂ 1b ₁	0.2887	0.4511
H ₂ O(R_e)	1b ₂ 3a ₁	0.0282	0.0342
H ₂ O(1.25 R_e)	1b ₂ 3a ₁	0.0408	0.0659
H ₂ O(1.5 R_e)	1b ₂ 3a ₁	0.0597	0.1001
H ₂ (R_e)	1 σ_g	0.0183	0.0057
H ₂ (1.25 R_e)	1 σ_g	0.0240	0.0104
H ₂ (1.5 R_e)	1 σ_g	0.0312	0.0160
CH ₄ (R_e)	2a ₁ 3a ₁ 1b ₁ 1b ₂	0.0259	0.0056
CH ₄ (1.25 R_e)	2a ₁ 3a ₁ 1b ₁ 1b ₂	0.0357	0.0438
CH ₄ (1.5 R_e)	2a ₁ 3a ₁ 1b ₁ 1b ₂	0.0488	0.0914

functions (VWN [12], LYP [13], P86 [18], P91 [19]). This idea is not new, and the quality of functionals such as BVWN, BLYP, BP86, BP91 is well known. It seems at least possible that if OPTX is a factor of two better than B88X, then a combination of OPTX with a quality correlation functional could be very attractive. We view LYP as a ‘semi-empirical’ functional, with its parameters fit to two electron atoms, which means that the dominant electron pair correlation energy is accounted for. VWN→P86→P91 represent improving correlation functionals which commence from the electron gas. Perdew would argue that they are not ‘semi-empirical’. We now denote OPTX by ‘O’ and we therefore examine the performance of the new exchange-correlation functionals OLYP, OP86 and OP91. We have a test deck of 93 atoms and molecules, for which we examine the performance for the prediction of atomic energies, molecular binding energies and molecular geometries. We have already examined [20] the performance of a number of functionals (including LDA, PBE [21], BLYP, BP86, BP91, HCTH [22] and hybrid functionals B3P86, B3LYP, B3P91). The new analysis with the new functionals is given in table 5, using the notation of [20].

As far as the energy analysis is concerned (atomic energies, ionization energies and binding energies) we see with d_E (the mean absolute error in kcal mol^{−1}), that OP91 is an improvement over BP91. A closer examination shows that the errors for individual molecules are rather different, and can be of a different sign, indi-

cating a continuing question on the accuracy of the P86 and P91 correlation functionals. OLYP offers a clear improvement from the energy analysis over the often used BLYP functional. An examination of the d_G values (which measure how much the gradient ∇E is non zero at experimental equilibrium), show that the geometrical predictions with the O functionals are a considerable improvement over the B functionals, and indeed the analysis suggests that the geometrical predictions of OLYP, OP86 and OP91 will be similar. Exchange, through its left-right correlation effects, is all important for structure prediction. Finally we observe that OLYP is truly competitive with B3LYP for energetics, and is becoming competitive for structures.

It is interesting to gather many of the results on N₂ together, and examine contributions to its binding energy (D_e), for which the exact value is 228.6 kcal mol^{−1}. We can separate contributions:

- computed HF definitive contribution: 114 kcal mol^{−1};
- computed left-right correlation contribution: B88X 54; OPTX 51, kcal mol^{−1};
- therefore dynamic correlation contribution: 60: 63, kcal mol^{−1}; and
- computed dynamic correlation contribution: LYP 68; P91 65, kcal mol^{−1}.

The values for the dynamic correlation energy computed with LYP and P91 are more consistent with the OPTX

Table 5. Performance of some exchange-correlation functionals over 93 atomic and molecular systems, examining mean absolute (d) and rms (r) energy errors (E) (in kcal) and energy gradient (G) (in E_h) quantities. The smaller the E quantities the better the energetic predictions, the smaller the G quantities the better the structural predictions.

Functional	d_E	d_G	r_E	r_G
BLYP	4.46	2.08	5.99	4.69
OLYP	3.26	1.45	4.11	3.09
BP86	6.74	1.85	8.45	3.82
OP86	6.70	1.42	8.16	2.77
BP91	4.72	1.74	6.09	3.66
OP91	4.21	1.33	5.38	2.65
B3LYP	3.40	0.91	4.42	2.49

exchange contribution than with the B88X exchange contribution.

This paper complements an earlier paper by one of us [23] where we examined the correlation energy for diatomic molecules. In that paper we looked at 8 molecules at equilibrium and stretched geometries, examining again the partition of the correlation energy into two parts. Exact values for the total correlation energy are deduced from HF and RKR values. In particular it gives values for the dynamic correlation energy, computed with the LYP and VWN functionals, and it is examined how well *ab initio* values agree with the DFT values. The earlier paper contains references to many of the original papers on the separation of the correlation energy into two parts.

7. Conclusion

The principal new work in this paper is that we report a new two parameter exchange functional, of the GGA form, which we believe is superior to the classic exchange functional of Becke, B88X. We have determined it by arguing that KS calculations on atoms with it should reproduce their HF energies. This functional, OPTX, was refined by least squares to the 18 first and second row atoms. Interestingly we found that inclusion of functional terms dependent on $\nabla^2\rho$ or $\sum_i(\nabla\phi_i)^2$ did not improve the refinement. Next we performed KS calculations on small molecules, and defined $E(\text{HF})-E(\text{OPTX})$ to be the left-right correlation energy. The value for this correlation energy differs from the corresponding value which is obtained using the B88X functional. The bond lengths predicted with B88X and OPTX are longer than observed. We have attempted to calculate the left-right correlation energy *ab initio* using the MCSCF procedure and some agreement is obtained for some molecules. Finally we have combined OPTX

with the correlation functionals LYP, P86, P91 to form OLYP, OP86 and OP91. An examination of the KS predictions for 93 systems with these functionals shows that structures are much improved when compared with BLYP, BP86 and BP91, and that energetics are much improved when OLYP is compared with BLYP. Indeed it is found that OLYP compares favourably with B3LYP. We think that OPTX is an improved exchange functional: so much so that if it was possible to determine a correlation functional of equal accuracy, then a combination of the two could be very useful. Finally we have underlined the point that an exchange functional necessarily introduces left-right correlation, and it is therefore inappropriate to compare HF calculations on molecules with DFT calculations using exchange-only functionals.

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